



Thapar Institute of Engineering & Technology

(Deemed to be University)

PATIALA

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Answer Sheet No.

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INSTRUCTIONS

1. Write on both sides of the answer-book.
2. Do not remove or tear off any page from this answer book!
3. Candidates found guilty of unfair means or misconduct shall be dealt with as per Institute rules on unfair means.
4. Write your Roll No. on the Question Paper.
5. Write Page No. of each question (which you have attempted) in the table given below.

Recheck $\rightarrow$ Q3(b)

Evaluation:

Signature of the Invigilator

Q.No.	I	II	III	IV	V	VI	VII	VIII	IX	X	Total Marks
Page No.											
Marks	6	5	5+5	0+1							

Total Marks in words:

Signature of the Examiner

Start Writing from here :-

Q1(a) func1:-

1st Loop worst case: Run = n 2nd Loop worst case: Run = n

3rd Loop worst case: Run = 10

So, Time complexity $\rightarrow n * n * 10$ $\rightarrow 10n^2$ $O(n^2)$ is worst time complexity.

func2:-

1st Loop worst case: Run = $2n$ 2nd Loop worst case: Run = $\frac{j}{3} = \frac{2n}{3}$ 3rd Loop worst case: Run = $j^2 = \left(\frac{2n}{3}\right)^2 = \frac{4}{9}n^2$ So, Time complexity $\rightarrow 2n * \frac{2n}{3} * \frac{4}{9}n^2$ $O(n^4)$ is worst time complexity

func3:-

1st Loop worst case: $\text{Run} = n$

2nd Loop worst case: $\text{Run} = n$ (i.e. $i=0$)

3rd Loop worst case: $\text{Run} = n$ (i.e. $i=j=0$)

So, Time complexity $\rightarrow n + n + n$
 $O(n^3)$ is worst time complexity.

Q1(b) Algorithm:-

① $\text{Min} = \text{arr}[0]$, $\text{count} = 0$

② for ($i=1$; $i < N$; $i++$) {

if ($\text{arr}[i] > \text{Min}$) {

$\text{Min} = \text{arr}[i]$;

$\text{cout} << \text{"Wait = " << ++count << endl; count = -1;$

} $\text{count}++$;

}

For $\text{rain}[] = \{20, 15, 25, 18, 9, 30\}$;

① $\text{Min} = 20$; $\text{count} = 0$.

② if ($15 > 20$) $\times$ $\text{count} = 1$

if ($25 > 20$) $\rightarrow \text{Min} = 25$, $\text{Wait} = 2$

$\text{count} = -1$

$\text{count} = 0$

③ if ($18 > 25$) $\times$ $\text{count} = 1$

if ($9 > 25$) $\times$ $\text{count} = 2$

if ($30 > 25$) $\rightarrow \text{Min} = 30$, $\text{Wait} = 3$

$\text{count} = -1$

$\text{count} = 0$

So, for following array worst wait time is 3 days.

Q2(a)

A	Stack	
60	60	60, 20
20	20	60, 20
2		60, 20, 2
1	1	60, 20, 2
+	+	60, 10
15		60, 10, 15
5		60, 10, 15, 5
*	+	60, 10, 15, 5
-	-	70, 75
100		70, 75, 100
+	+	70, 25
		70 + 25

Expr = 95

Q2(b)

(i) After 1st iteration, $Q_1 = [5, 3, 6, 4, 3]$ Rear1 = 0, Front1 = 1
 $Q_2 = []$ Rear2 = Front2 = -1

After 2nd iter, $Q_1 = [3, 6, 4, 3]$ Rear1 = 0, Front1 = 2
 $Q_2 = [5]$ Rear2 = 0, Front2 = 0

After 3rd iter, $Q_1 = [4, 3, 3]$ Rear1 = 1, Front1 = 4
 $Q_2 = [5, 3]$ Rear2 = 1, Front2 = 0

After 4th iter, $Q_1 = [3, 3, 5]$ Rear1 = 2, Front1 = 0
 $Q_2 = [5, 3]$ Rear2 = 1, Front2 = 0

After 5th iter, $Q_1 = [5, 3]$ Rear1 = 3, Front1 = 2
 $Q_2 = [5, 3, 3]$ Rear2 = 2, Front2 = 0

(ii)

After 1st iteration : $Q_1 = [3, 5, 3, 6, 7]$
 $Q_2 = []$

Rear1: -3 ; Front1: -
 Rear2: -1 , Front2: -1

After 2nd iter: $Q_1 = [3, 2, 5]$
 $Q_2 = [3]$

After 3rd : $Q_1 = [3]$
 $Q_2 = [3, 3]$

②

After 4th : $Q_1 = \text{empty}$
 $Q_2 = [3, 3, 3]$

Rear1 = Front1 = -1
 Front2 = 3 , Rear2 = 0

Q3(a) Node * head 3;
 Node * ptr3 = head 3;
 Node * ptr1 = head 1;
 Node * ptr2 = head 2;

loop:-

while (ptr1 -> next != null && ptr2 -> next != null) {

if (ptr1->value > ptr2->value) {

int* ptr3 = new nn;

nn->value = ptr1->value;

nn->next = ptr2;

nn->next->value = ptr2->value;

nn->next->next = ~~ptr3~~ head 3; null;

}

else {

nn->value = ptr2->value;

nn->next = ptr1;

nn->next->value = ptr1->value;

nn->next->next = null;

}

}

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① We compare value of ptr1 & ptr2. Then insert inside new Linked List 3 through ptr3.

② Loop run until ptr1 -> next or ptr2 -> next != ~~head~~ null. Then we add remaining values of other directly to LL3.

③ We are giving ptr3 -> next = null each time. So, that it is easy to transverse in next step through

while (ptr3->next != null) ptr3 = ptr3->next;

Q3(b) First we find position by Binary Search.

① $Start = 0$, $end = n-1$, $mid = start + end/2$

```

② BinSrch(int arr[], int x) {
    while (start < end) {
        if (arr[mid] ≤ x) {
            if (arr[mid+1] ≥ x) {
                return mid;
            }
        }
        if (mid > key) {
            end = mid - 1;
        }
        else {
            start = mid + 1;
        }
    }
}

```

We got position, the value should be placed after $arr[mid]$.

Now, we make space by shifting values after mid.

```

③ for (i = n; i > mid; i--) {
    arr[i+1] = arr[i];
}

```

④ Now, give value
 $arr[mid+1] = x;$

Time complexity $O(\log n)$ for binary search (worst case).

Q4(a)

① 45, 36, 10, 12, 78

② 36, 10, 12, 45, 78

③ 10, 12, 36, 45, 78

Sorted

No Algorithm is stable and best always it depends on case. As, in selection sort first we need to find max value in array. If value is at end then it take too much time in loop.

Q4(b)

1) maxNode $\rightarrow$ amount $>$ curr $\rightarrow$ amount2) minNode $\rightarrow$ amount $<$ curr $\rightarrow$ amount

3) curr

4) head $\rightarrow$ next5) curr $\rightarrow$ next $\neq$ null

6) return

7)

8)

9) return

10)